ADITYA BAYU PRATAMA

Lead Software & IT Engineer | Smart Manufacturing & Industry 4.0 Specialist

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PROFESSIONAL PROFILE

Results-driven **Lead Software & IT Engineer** with 7+ years of expertise in Manufacturing Execution Systems (MES), Industrial IoT, Machine Vision, and Smart Factory Automation across high-precision electronics and oil & gas sectors. Proven track record of scaling IT teams, architecting custom C#/.NET enterprise solutions for 500+ product models, and executing high-stakes Industry 4.0 digital transformations. **2-Time 1st Place Winner of the Giken Global Value Engineering Award (2023 & 2025)**. Adept at bridging hardware, enterprise software, and AI-driven automation.

CORE COMPETENCIES & TECHNICAL SKILLS

Software & Languages:	C# (.NET Framework), SQL (MySQL), Python, React.js, C++, LabVIEW, Dart/Flutter, REST APIs
Smart Manufacturing:	MES Architecture, Industrial IoT (IIoT), Pokayoke System, AOI Machine Vision, Thermal Printer Automation
Hardware & Automation:	PLC (BNSP Certified), Raspberry Pi, Arduino, Lidar & Laser Sensors, Data Acquisition (DAQ) Systems
Engineering Quality:	FMEA, DMAIC, Six Sigma Practices, 5M Change Management, Continuous Improvement (Value Engineering)
IT & Leadership:	Team Recruitment & Management (Led 5 Engineers), Network Infrastructure, Enterprise Server Maintenance

HONORS & GLOBAL AWARDS

🏆 1st Place – Giken Global Value Engineering (2025)
Awarded top global prize for developing *Intelligent Machine Data*, optimizing real-time analytics & line efficiency.

🏆 1st Place – Giken Global Value Engineering (2023)
Won 1st place in inaugural entry with *Manufacturing Traceability System*, eliminating defect passage across lines.

PROFESSIONAL EXPERIENCE

GIKEN PRECISION INDONESIA — Batam, Indonesia

Lead Software & IT Engineer (Promoted from IT Engineer / Team Lead) — *Apr 2019 – Present*

- **Enterprise MES System Architecture (C# & SQL):** Designed and deployed a robust, custom Windows-based Manufacturing Execution System currently governing over 500 product models across Automotive, Medical, and Consumer Electronics lines, enforcing Pokayoke defect prevention and CTQ quality compliance.
- **Digital Transformation Web Platform (React.js / MySQL):** Built an enterprise-wide web application digitizing legacy manual workflows, including employee records, departmental org charts, overtime request approvals, ticketing tracking, and project scheduling.
- **Industrial IoT Gate & ESD Safety Automation:** Engineered an automated ESD wristband monitoring auto-gate using Arduino, C#, and MySQL, enforcing 100% compliance before line entry and tracking hardware defect rates.
- **100% Automated Thermal Label Printing Pipeline:** Streamlined QR-code label printing workflows across new electronic lines by building a direct C# and SQL interface to thermal printers, eliminating printing delays.
- **Centralized Data Storage & Infrastructure Security:** Established centralized network folder sharing servers across all departments, preventing hard drive failure data loss and boosting cross-departmental collaboration.
- **IT Team Leadership & Department Growth:** Recruited, mentored, and scaled the software and IT engineering department from a solo role to a high-performing 5-engineer team, managing NPI testing and digital infrastructure.
- **Mission-Critical Incident Response & Problem Solving:**
  - *Worm Virus Outbreak Mitigation (2026):* Successfully contained and eliminated a malware outbreak within 3 hours across 20 compromised laptops, restoring drive file structures without production downtime.
  - *High-Availability Server Diagnostics:* Diagnosed complex hardware kernel and power management faults on core MySQL servers, achieving stable 99.9% uptime.
  - *Network SLA Management:* Rapidly resolved critical factory network outages under 15-minute strict SLAs using systematic line isolation procedures.

CLADTEK BI-METAL MANUFACTURING — Batam, Indonesia

Junior Programmer – R&D Department — *Apr 2018 – Jan 2019*

- **Autonomous Welding Defect Inspection Robot:** Engineered a pipe-inspection robot using C#, Raspberry Pi, Lidar, and specialized light sensors to autonomously navigate inside industrial pipes, streaming real-time video and calculating defect locations and severity.
- **High-Speed Data Acquisition Protocols:** Developed low-latency data transmission pipelines connecting embedded hardware sensors to central servers for real-time welding QA evaluation.

SELECTED TECHNICAL & AI PROJECTS

- **AI-Driven Grid Trading Application (Crypto & Stocks):** Developed a high-frequency trading bot in React.js and Telegram Bot API using zero-reasoning AI models for ultra-low latency execution based on market sentiment, technical indicators, and Binance API integration.
- **LabVIEW AOI PCB Inspection System:** Developed an Automated Optical Inspection (AOI) algorithm in LabVIEW Vision to detect missing or misaligned components on PCB assembly lines, significantly reducing manual QA headcount and ST production time.
- **IoT Emergency Response & Human Finder Robot:** Built a Raspberry Pi/Arduino-powered wireless rescue robot equipped with thermal, gas, smoke, PIR sensors, and live camera streaming controlled via web/mobile apps for hazardous environment search.
- **Smart Multi-Biometric Access Controller:** Built an automated door lock ecosystem using Python (OpenCV) for facial recognition, fingerprint validation, RFID, and PIN passcodes backed by MySQL logging.

EDUCATION & PROFESSIONAL CERTIFICATIONS

Bachelor of Applied Science (S.Tr.T.) - Mechatronics Engineering

State Polytechnic of Batam | *Graduated Cum Laude (GPA: 3.81 / 4.00)*

Key Focus: AI & Robotics, Industrial IoT, Machine Vision, Embedded Systems

Key Certifications:

- **Google Data Analytics Professional** (Google/Coursera)
- **PLC Competency Certificate** (BNSP Indonesia)
- **Electrical Power Technician Class II** (LPJK)
- **DevOps & Network Engineering** (Dicoding Academy)